

COSC 1336 – Fundamentals of Programming

Project Assignment #8 - Student Database

Objective

Write a Python program that creates and manages a **student database** using a **dictionary**. The dictionary will use the **student ID** as the key and the **student name** as the value.

Functional Requirements

1. Data Input:

- The program will read from a **text file** (e.g., data.txt) that contains student records.
- Each line in the file will be in the format:
studentID,studentName
Example:
A007,Ally Baba
B123,He Man
A123,Super Man
C234,Spider Man

2. Dictionary Creation

- The program will parse the file and populate a **dictionary** where:
 - studentID → dictionary key
 - studentName → dictionary value

3. User Interaction

- The program will repeatedly prompt the user to enter a **student ID**.
- If the ID is found in the dictionary, display the corresponding **student name**.
- If not found, display: **"Student Not Found"**.
- The program continues until the user enters **'N'** (not case-sensitive) to stop.

Required Functions

You **must use** the following functions in your program:

- getStringData()
 - Prompts the user for input.
 - Validates that the input is non-empty.
 - For exit prompts, ensure the return value is the uppercase form of 'Y' or 'N'.
- readData()
 - Opens and reads the file.
 - Parses each line and populates the dictionary.
 - Returns the dictionary.
- displayResult(studentID, studentName)
 - Accepts the student ID and the corresponding name (or error message) and displays the result.
- main()
 - Controls program flow: calls all other functions and manages input/output.

Additional Guidelines

- Validate user input (no empty strings).
- Follow professional programming practices (naming conventions, indentation, documentation).
- Include comments at the beginning of the file and before each function.

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Sample Output
(User entry is in RED)

Project #8
Written by: Ally Baba

Search Student Information

Are you ready to search? (Y/N) **Y**
Please enter a student ID: **A007**

Query Summary: Student Information

Student Found
Name: Ally Baba
ID: A007

Are you ready to search? (Y/N) **Y**
Please enter a student ID: **X007**

Query Summary: Student Information

Student Not Found
Are you ready to search? (Y/N) **N**
You have searched 2 times

End of Project 8